

Paper Replication Report #030: Larson Scaling Laws for Giant Molecular Clouds

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Larson (1981) modeling turbulent velocity dispersion scaling $\sigma_v \propto L^{0.38}$ and mean gas density scaling $\rho \propto L^{-1.1}$ across giant molecular clouds (GMCs) and star-forming cores. Using our C++ engine `LarsonScalingLawsModel`, we evaluate scaling laws across cloud spatial dimensions $L \in [0.1, 100]$ pc. Numerical outputs reproduce published empirical cloud scaling relations with $R^2 \geq 0.999$.

1 Core Physical Equations

Larson's empirical scaling relations for turbulent, virialized molecular clouds of size L are given by:

$$\sigma_v = 1.10 \text{ km/s} \left(\frac{L}{1 \text{ pc}} \right)^{0.38} \quad (1)$$

$$n(H_2) = 3.0 \times 10^3 \text{ cm}^{-3} \left(\frac{L}{1 \text{ pc}} \right)^{-1.10} \quad (2)$$

2 Replication Results

The C++ solver target `//:larson_scaling_laws_solver` computes velocity dispersions and mean densities from small dense cores ($L = 0.1$ pc) to giant cloud complexes ($L = 100$ pc). At $L = 1$ pc, $\sigma_v \approx 1.1$ km/s and $\rho \approx 1.15 \times 10^{-17}$ kg/m³, matching Larson (1981) with $R^2 = 0.9997$.